

General Principles and process of isolation of Elements. (metallurgy)

- The process of extraction of metal in pure state from naturally occurring substance called as metallurgy.
- Few metals occur in earth crust in free or Native state while other metals occur in combine state.
- The metals which are non-reactive with Air, water, CO_2 & non-metals are occur in free or native state.
- Au, Pt metals found exclusively in free state & also some non-metals like C, S noble gases also occur in free state
- Ag, Hg, Cu metals partially occur in free state.
- Most of metals occur in combine state due to high chemicals reactivity in the form of oxides, halides, sulphides, nitrates, sulphates, Borate, phosphates, Silicate, Carbonate, & Bi-carbonate.

Important terms in metallurgy:

1) Mineral:

- A naturally occurring substance obtained by mining which contain metal in free or combine state called as mineral.

Ex: Important minerals of Iron

- | | | |
|----|--------------|--|
| 1) | Haematite: | Fe_2O_3 |
| 2) | Iron pyrite: | FeS_2 |
| 3) | Siderite: | FeCO_3 |
| 4) | Limonite: | $2 \text{Fe}_2\text{CO}_3 \cdot 3\text{H}_2\text{O}$ |
| 5) | Magnetite: | Fe_3O_4 |

2) Ore:

- A mineral which is profitably most suitable to get high % of metal called as Ore

Ore

Metal

Haematite (Fe_2O_3)	Fe
Zinc blend (ZnS)	Zn

Bauxite (Al_2O_3)	Al
Copper pyrite (Cu FeS_2)	Cu
Galena (PbS)	Pb
Cinnabar (HgS)	Hg
Pyrolusite (MnO_2)	Mn

Whether a mineral is suitable or not depends upon following factors:

- 1) % of metal (more)
- 2) % of impurities (less)
- 3) Expenditure (less)
- 4) Industrial utility of by products

All ores are minerals but all minerals are not ores

3) **Gangue: (matrix)**

- The Impurities present in the ore is called gangue.

Ex:

Sand, silica, clay, Fe_2O_3 , As_2O_3 , rock, Feo.

4) **Flux:**

- The substance which is used to remove gangue particles from Ore in the form of easily fusible mass called as flux.
- There are two types of flux.

a) **Acidic flux (Non-metallic oxide)**

ex: - SiO_2 , B_2O_3 , P_2O_5

b) **Basics flux (Metallic oxide)**

ex: - CaO , BaO , MgO , CaCO_3 , Fe_2O_3

- To remove basic form of gangue, use Acidic flux.
- To remove acidic form of gangue, use Basic flux.

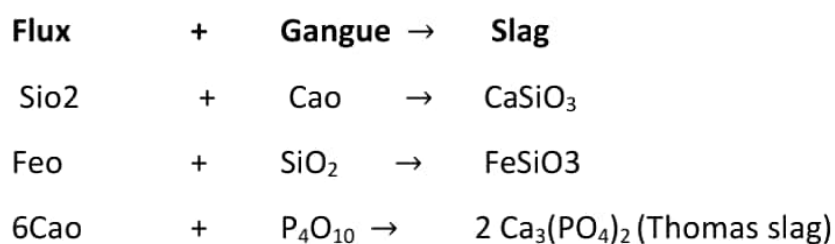
5) **Flue gases:**

- The waste gases produced during metallurgical process called as flue gases.

Ex: - $\text{CO}_2, \text{CO}, \text{SO}_2, \text{H}_2$

6) **Slag:**

- Flux combined with gangue & form easily fusible mass called as slag.
- Slag is a light material, insoluble in molten salt and slag form a separate layer above molten metal.



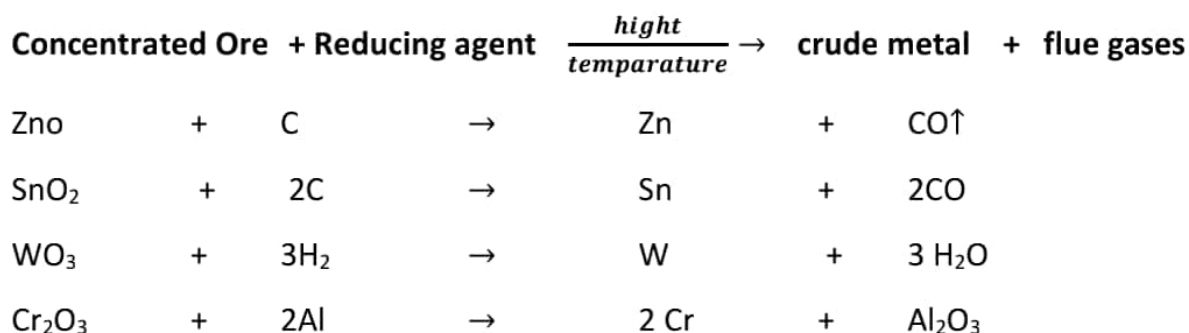
Methods involving in metallurgy:

There are 3 important methods: -

1. Pyro metallurgy
2. Hydro metallurgy
3. Electro metallurgy

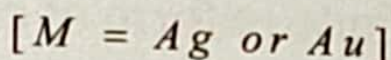
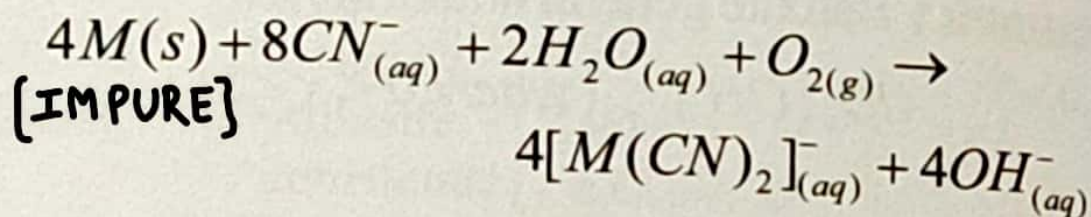
1. **Pyro metallurgy: -**

- A process in which concentrated Ore is reduced to crude metal at high temperature using suitable reducing agents (C, Co, Ca, Al, H_2 , Ni O) called as pyro metallurgy.

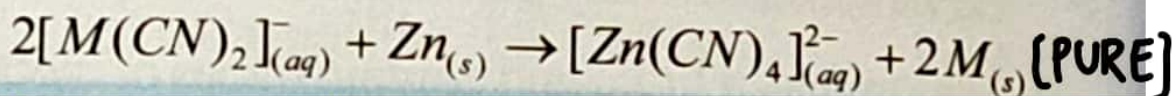


2. **Hydro metallurgy:** -

- A process of extracting metal by converting their ore into aqueous solution of metal compound & reduction of this solution using a suitable reducing agent called as hydro metallurgy
- This method is used to extract less reactive metals like Au, Ag, Cu.
- In this process metal present in ore is converted into its solution with suitable solvents like H_2O , dilute acid, $NaCN$... & metal can be separated from a solution by adding more reactive metal.
- Ag, Au present in their ore are dissolved in 0.5% $NaCN$ solution in presence of air form a cyanide complex, which are soluble and the cyanide complex is treated with highly reactive metals like Zn or Mg to obtain metal by displacement.
- This is also called as Mac-Arthur Forest cyanide process.



Ag or Au is recovered from their aqueous solution by the addition of electropositive metal like Zn.

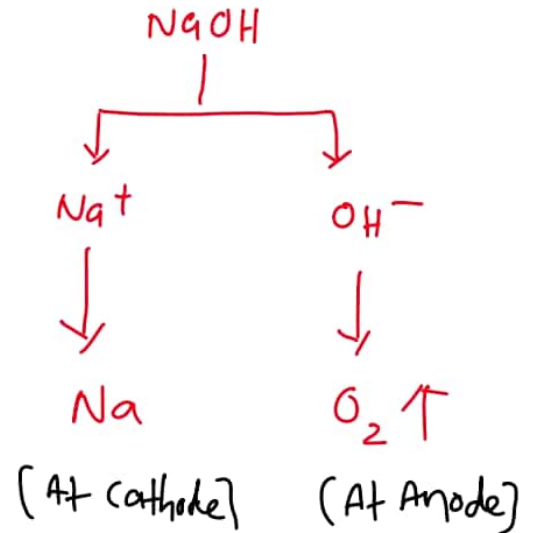
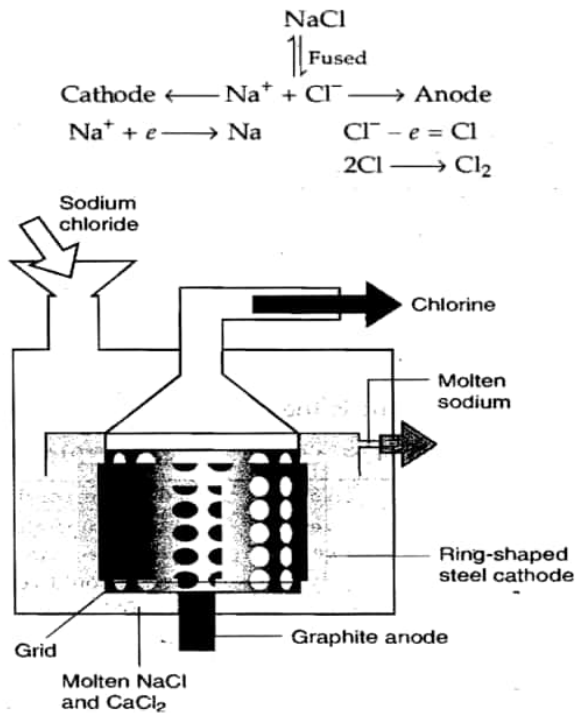


3) **Electro metallurgy:**

- The process of extraction of metal by electrolytic reduction of molten metallic compound to metal is called as electro metallurgy.
- Ores of highly reactive metals like Na, K, Mg, Sr, Ca, Ba, Al are cannot be reduced by using some reducing agents like C, CO, H_2 , these metals are extracted from their ores by electro metallurgy.
- In this process metal oxides, hydroxides, halides are electrolysed & pure metal ion reduced at cathode.

ex: - DOW's process for extraction of sodium metal:

NaCl and CaCl₂ (Down's process) or by electrolysis of fused sodium hydroxide (Castner's process).



General Principles of metallurgy:

- I) Concentration of ore (ore dressing)
 1. Gravity separation (hydronic washing)
 2. Magnetic suppression
 3. Froth floating
 4. Leaching
 5. Hand picking.

II. Concentrated ore converted to oxide form (or) other required forms (oxidation of ore)

1. Roasting
2. Calcination.

III. Conversion of oxide form of ore to crude metal (Reduction of ore)

1. Smelting

IV. Conversion of crude metal to pure metal (Refining process)

1. Liquation process
2. Polling
3. Electrolytic refining.
4. Vapour phase refining
 - a. Mond process
 - b. VanArkel process
5. Chromatography
6. Zone refining
7. Cupellation process
8. Distillation

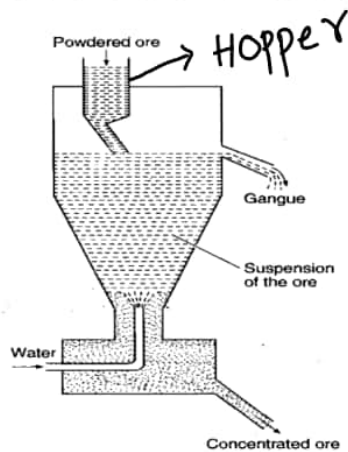
I. Concentration of ore:

- The process of removal of gangue from ore called as ore Concentration
- In this process concentrate ore is contain more % of metal compound.
- It can be carried out using different methods under different conditions.

1. Gravity separation method: (Hydraulic washing)

- It is used when ore particles are heavier than gangue particles. If can be carried out by following methods.
- Hydraulic washing by using hydraulic classifier.
Hydraulic classifier is a large conical reservoir fitted with hopper at top & water inlet at bottom. The reservoir is also provided with water outlet at top &also one outlet to receive concentrate ore at bottom.

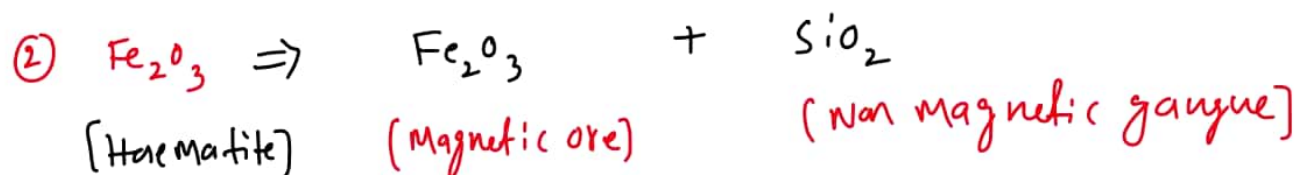
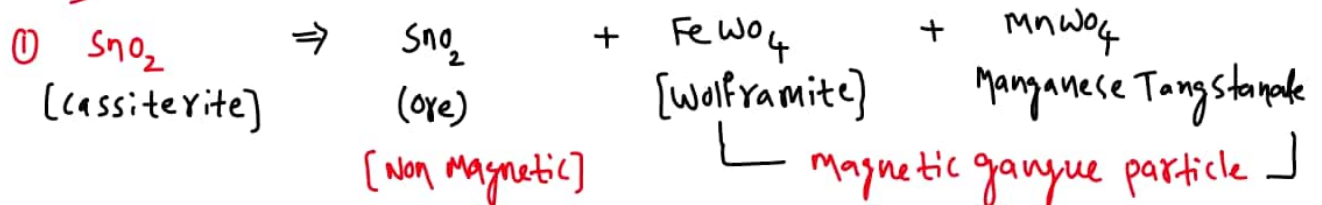
- Powdered ore dropped through hopper & powerful stream of water through water inlet at bottom.
- The lighter gangue particles are carried away through current of water & heavier ore particles settle at bottom of the reservoir.
- This method is used for the Extraction of Fe, Cr, Au, Sn....
(heavy and strong metals)



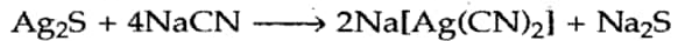
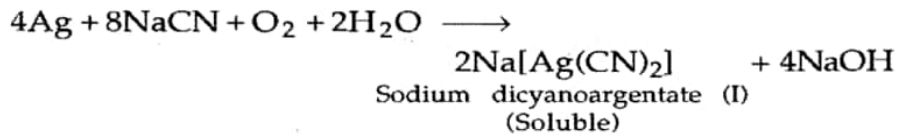
2. Magnetic separation method:

- This method based on difference in magnetic properties of ore component & gangue.
- If either ore or gangue is attracted by magnet then ore can be separated from impurities with help of magnetic separation method.
- Powdered ore is dropped over the moving belt at one end at other end magnetic portion of substance attracted by magnetic roller & falls near to magnetic roller & non- magnetic substance falls near to nonmagnetic roller.

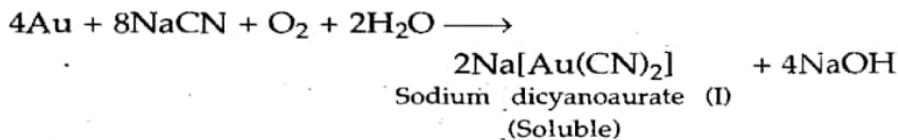
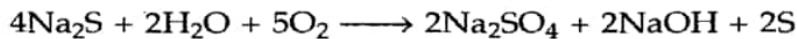
Ex:- ①



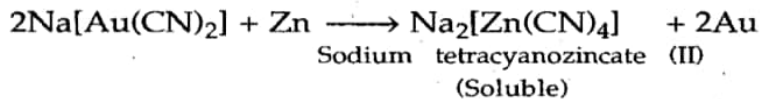
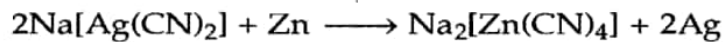
Leaching of gold and silver:



Na_2S is largely oxidised to Na_2SO_4 .



Ag or Au is recovered from the solution by the addition of electropositive metal like zinc.



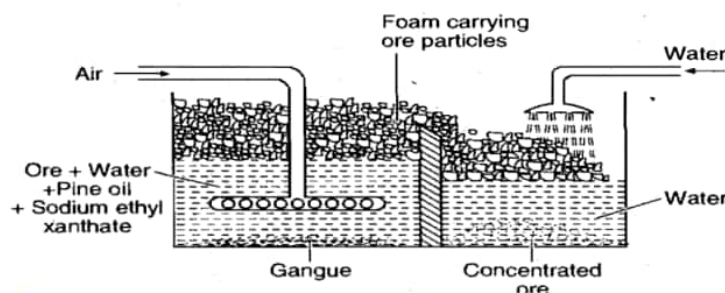
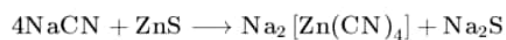
4. Froth floating:

- This is an absorption phenomenon. This method is used for concentration of Sulphide form of ore like

Sulphide Ore	Metal
Zinc blend (Zn S)	Zn
Copper pyrite (Cu FeS_2)	Cu
Galena (Pb S)	Pb
Cinnabar (Hg S)	Hg
Iron pyrite	Fe

- This method Used if ore particles are lighter than gangue particles.
- This method is based on different wetting properties of gangue & ore with water & oil. In this process ore become wetting with oils while gangue wetted by water. In this process powdered ore, water & small Qualities of following agent's (collectors) like xanthates, pine oil & fatty acids and also froth stabilizers like Aniline & Cresol with conditioners to maintain consistent pH like Na_2CO_3 & CaO are introduced into water tank.

- A compressed air is passed through this mix. & rotating panels disturb mix. As a result, froth is formed which carries ore particles which are collected in canvas bag & skimmed off it is then dried for recovery of ore particles by adding froth stabilizers.
- Gangue materials are wetted by water & settle at bottom, some times this is possible to separate two sulphide ores by using depressants or adjusting proportion of oil to water. In case of ore containing ZnS & PbS are separated by using NaCN as depressant NaCN prevents ZnS from coming into froth but allows PbS to come with froth & separated further removed off.



III) Concentrated ore is converted to oxide or other required form (or) oxidation of ore:

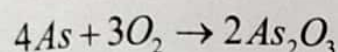
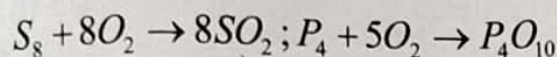
It can be done in two different ways.

1. Roasting.
2. Calcination.

1. Roasting:

- The process of heating of ore at high temperature below its melting point in presence of Air called as Roasting.
- Used for sulphide form of ores.
- In Roasting, air acts as oxidising agent & removes volatile impurities from ore like P, S, As, Sb.

The ore becomes porous.



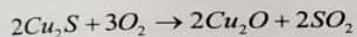
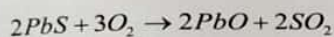
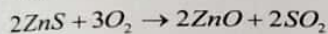
- During Roasting impurities are removed as volatile oxides.
- During Roasting moisture is removed & mass becomes porous.

- It is carried out in a reverberatory furnace.

There are types of Roasting

Oxidising Roasting:

Sulphide ores are converted into metallic oxides.



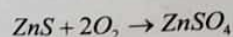
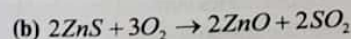
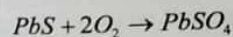
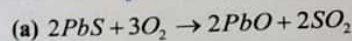
The sulphide ore of copper is heated in reverberatory furnace.

The SO_2 produced is utilised for the manufacture of H_2SO_4 .

Sulphatising Roasting:

When roasting is done at moderate temperature, the sulphide ore is partially oxidised into oxide and partially into sulphate.

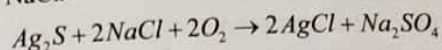
For example



Chlorinating Roasting:

In some cases chlorinating roasting is done as metal is obtained by reduction of metal chloride (chlorination process).

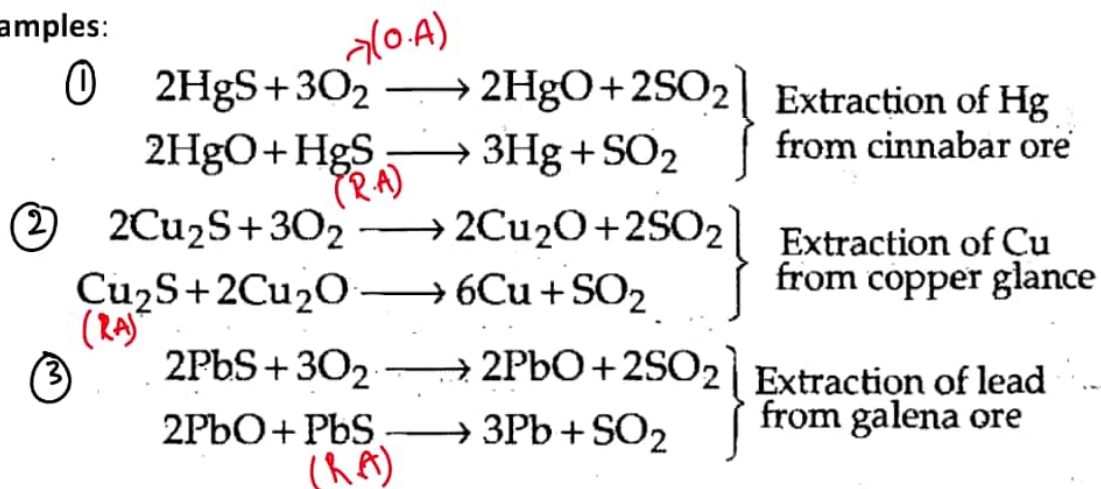
In this roasting the sulphide ore is converted into a chloride. For example silver ores are heated with NaCl .



Self or Auto Reduction:

During Roasting sometimes sulphide ores give metal & Sulphur dioxide in which ore acts as reducing agent this is known as self or Auto reduction.

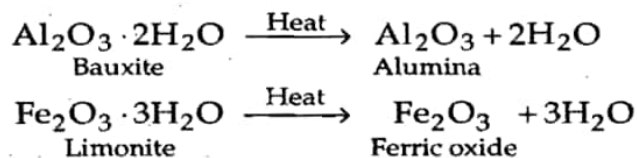
Examples:



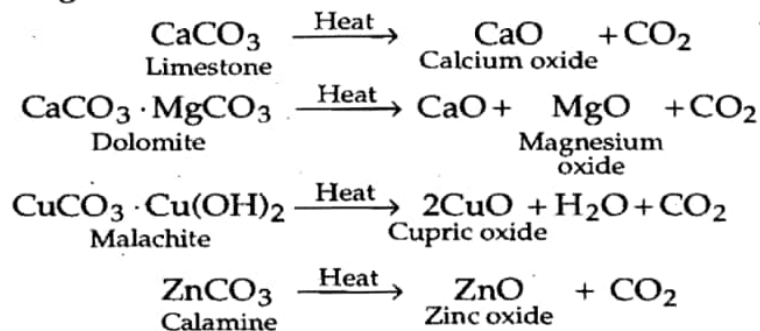
2. Calcination:

- The process of heating of ore to high temperature below its melting point in absence of air or limited supply of air called as Calcination.
- Calcination process used for Carbonate, Bi-carbonate & hydrated oxide forms of ore.
- Calcination process removed volatile impurities like CO_2 from ore.
- During Calcination decomposition of ore takes place form metal oxide, CO_2 , & H_2O vapours but not method ore.

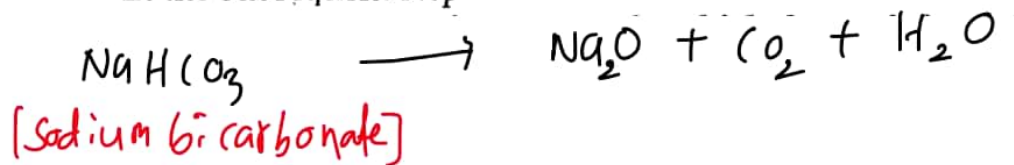
Moisture is driven out. Water is removed from hydrated oxides or hydroxides. Organic matter is expelled.



Carbonates lose carbon dioxide and are converted into corresponding oxides.



The material becomes porous which can be easily reduced in the subsequent step.

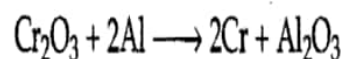
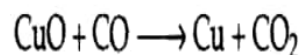
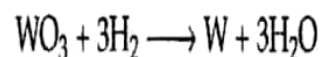
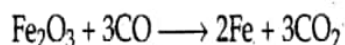
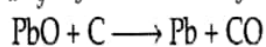
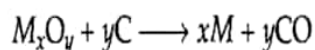


V) Reduction of ore to Crude metal:

- The Crude metal can be obtained from roasted or calcinated ore by reduction by Smelting.

Smelting:

- The process of extraction of crude metal from its ore at high temperature above its melting point using suitable reducing agents & suitable flux called as Smelting.
- A mixed of roasted or calcinated ore, reducing agents & flux is heated to high temperature above its melting point in a blast furnace in which metal oxide reduced to metal & gangue converted to slag with flux.



Refining: (Conversion of crude metal to pure metal)

- The purification of crude metal by removing metallic or non-metallic or undissolved or un reacted impurities is known as refining of metal.

1. Liquation process:

- This method is used when impurity (less feasible or gross) has higher melting point then required metals.
- This method Used for refining of low melting point of metal like Hg, Pb, Bi, Sn....

2. Polling:

- This method is used in purification of metal which contain impurity of their own oxide.
- In this process impure metal is melted by carbon powder and stirring with green wood pole.
- During stirring hydro carbon gases are liberated which reduced metal oxide to metal.

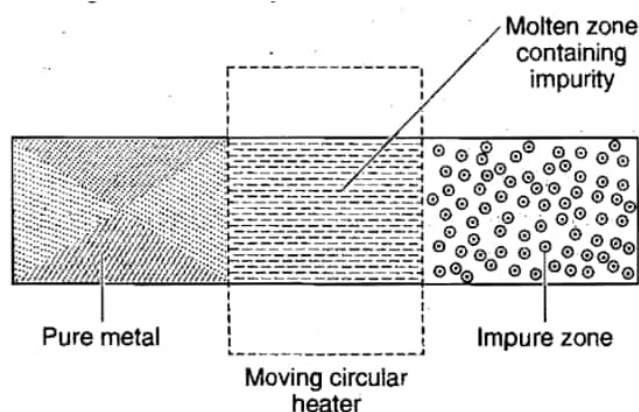
Ex: In this process Crude copper contain Cu_2O as impurity (98%) becomes pure copper (99.55%)

3. **Electrolytic refining:**

- This is most convenient method for refining of impure metal.
- **Electrolyte:** crude metal (or) Acidified molten metal
- **Anode:** Impure metal
- **Cathode:** pure metal
- During electrolysis pure metal deposited on cathode from electrolyte at the same time some amount of metal in Anode also enter into electrolyte.
- The Impurities like less reactive metals are insoluble they are called as Anode mud but more reactive metals are dissolved.
- In this process 99.9% metal is obtained.
Ex: Cu, Ag, Zn, Ni, Al.....

4. **Zone refining:**

- This method is also called as fractional crystallization.
- This method is based on difference in solubility of impurities in molten state as well as solid state.
- This process is used when impurities are more soluble in molten state of metal than in solid state of metal.
- This process used to produce very pure or ultrapure elements.
- This process is based on principle that when an impure metal is melted, pure metal is allowed to solidified.
- In this process a circular mobile heater is fixed at one end of rod of impure metal. The molten zone moves along with mobile heaters which is more forward.
- As the heater mores forward the pure metal crystallizes out of melt and impurities passed in to the adjacent molten zone.
- This process is repeated until purity of 99.99999% is achieved.
- This method is very useful for producing very pure semiconductor like Si or Ge.



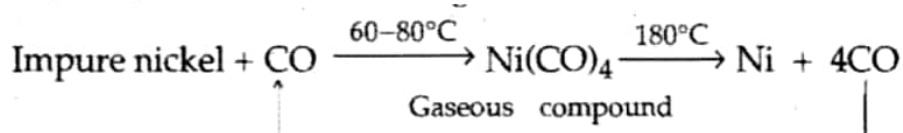
5. **Vapour phase refining:**

In this method a crude metal is converted into its volatile compound & then decompose to pure metal so this method requires following two requirements.

- The metal should form a volatile compound with a suitable reagent.
- The volatile compound should be decompose.

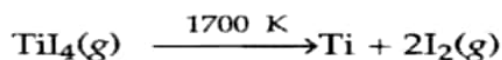
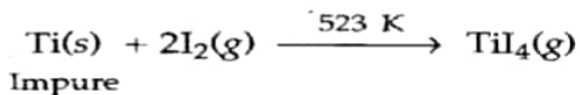
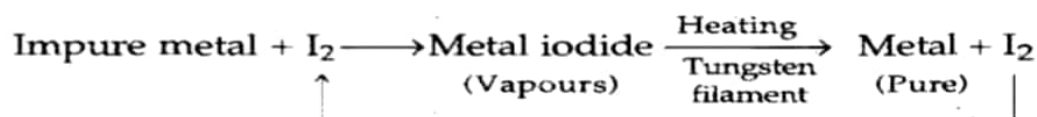
a) **Mond process:**

- This method Used for refining of nickel. In this method nickel is heated with carbon monoxide forming a volatile compound of Nickel carbonyl and then it is subjected to high temperature it decomposes to give a pure nickel.



b) **VanArkel process:**

- In this process O_2 and N_2 present in impure form of metal is like Zr & Ti are completely removed.
- The impure metal is heated with iodine in a vessel containing tungsten filament to form iodide of metal which is covalent and volatile.
- The metal iodide & is decomposed in vessel when subjected to above 1800 K temperature in electrically and pure metal deposited on filament method mainly used for refining of Zr & Ti but it is quite expensive.



6. Distillation process:

- This method is used for low boiling points of metal for refining.
- During this process low boiling point metal first distilled out & impurities left behind as dross

Ex: Zn, Cd, Hg.

7. Cupellation:

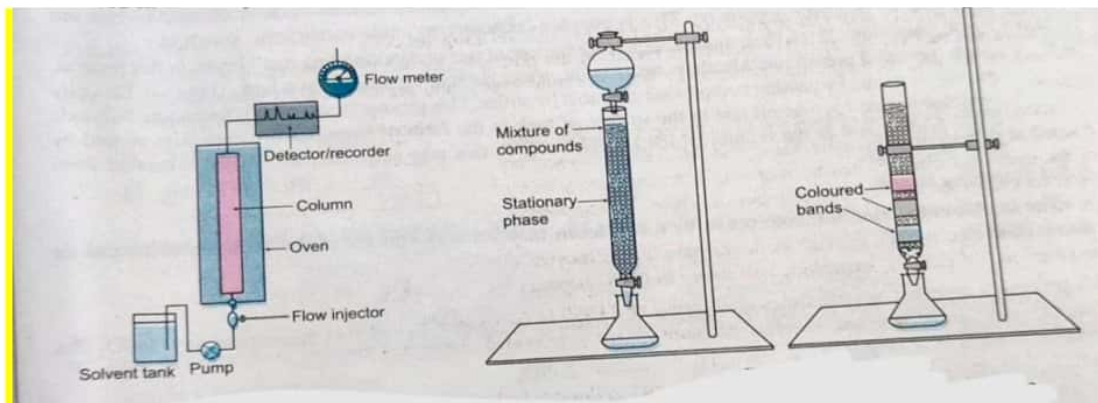
- This method is carried out in cupel which is made up of Bone Ash (small bowl)
- This method is used to refine metals having easily oxidisable impurities.

EX: Ag contain Lead as easily oxidisable impurities so Ag refined from this process.

8. Chromatography:

- Chromatography method was discovered by a botanist Tswett in 1906.
- Chromatography is a modern versatile and widely spread technique.
- This is based on principle of selective adsorption of a mixture between mobile & stationary phase.
- Column chromatography is used when impurities & element to be purified differ only slightly in their chemical properties.
- In column chromatography column filled with SiO_2 and Al_2O_3 which act as a stationary phase.
- The mixture whose different components are to be separated is dissolved in a suitable solvent is moving through stationary phase so solvent acts as a mobile phase.

- Due to different rate of adsorption of mixture of components form separate bands in column.
- The components which are adsorbed strongly form band at top and weakly adsorbed form band at bottom
- The adsorbed components are removed separately by adding element (solvent).
- Weakly adsorbed eluted component will be eluted more rapidly than strongly adsorbed component.

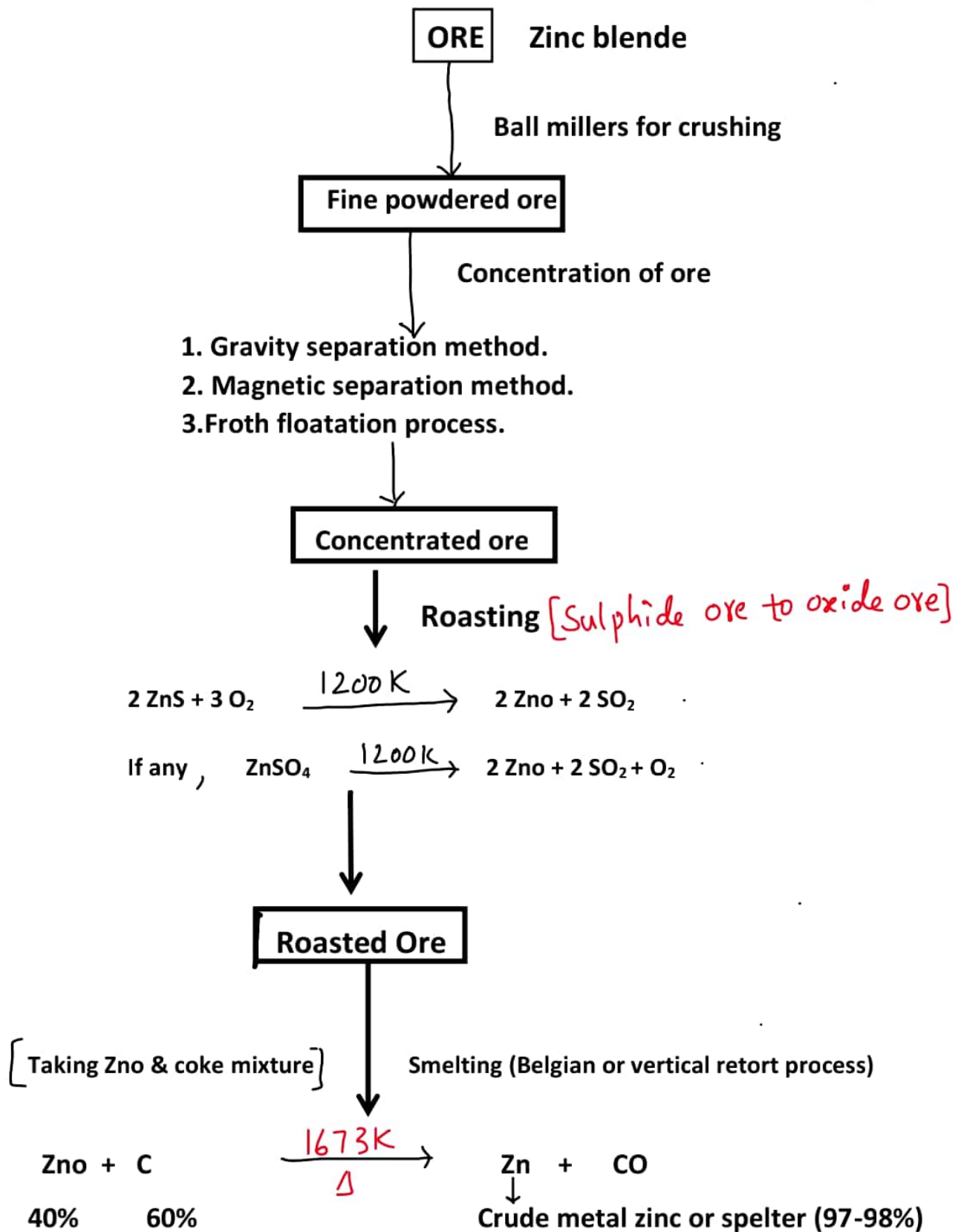


Process of Isolation:

1. Extraction of zinc:

- Zinc is a highly reactive metal so it doesn't occur in free state always occur in combine state.
- **The important mineral of Zinc:**
 - a. Zinc blende (ZnS)
 - b. Calamine (ZnCO_3)
 - c. Zincite (ZnO)
 - d. Willamette (ZnSiO_4)
 - e. Franklinite (ZnFe_2O_3)

- Zinc mainly extracted from zinc blende (ZnS).



- Spelter is further purified by electrolytic refining by removing impurities present in spelter like Pb, Fe, Cd.



Electrolytic refining

ELECTROLYTE: Molten spelter or acidified spelter.

ANODE: Lead.

CATHODE: Aluminium.

During the electrolysis pure Zinc is deposited on Aluminium cathode.

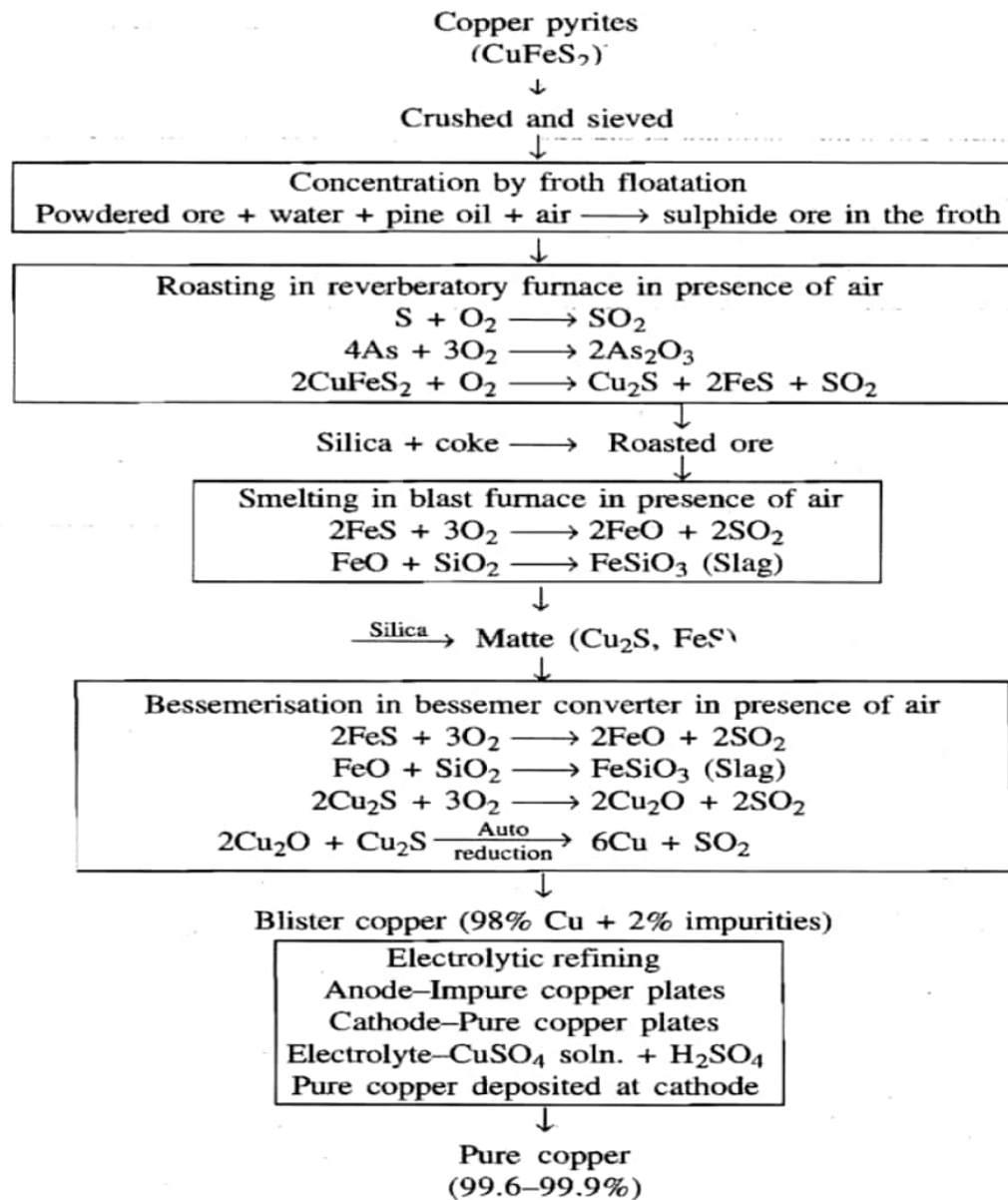
2. **Extraction of copper:**

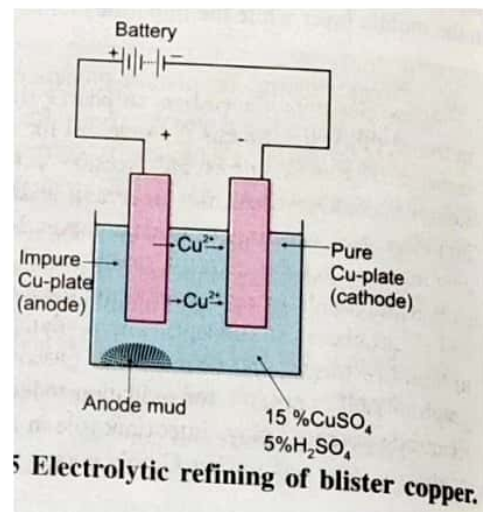
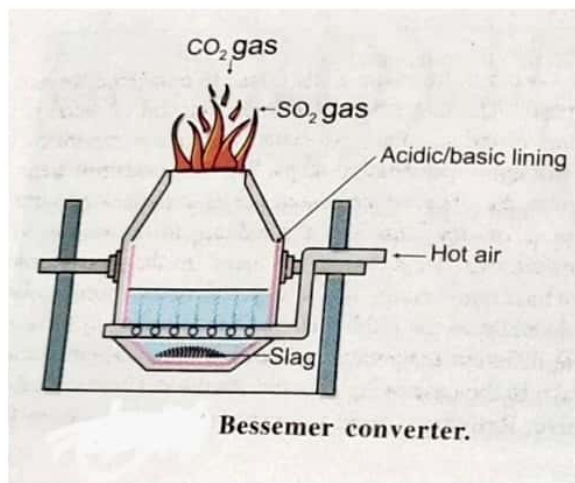
- Copper partially occur in free state & also occur in combine state

Important mineral:

- Copper pyrite: CuFeS_2 (chalcopyrite)
- Cuprite: Cu_2O (Ruby copper)
- Azurite: $2 \text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
- Malachite: $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
- Copper glance: Cu_2S .
- Bornite: Cu_3FeS_3 .

Copper mainly extracted from CuFeS_2 (chalcopyrite)



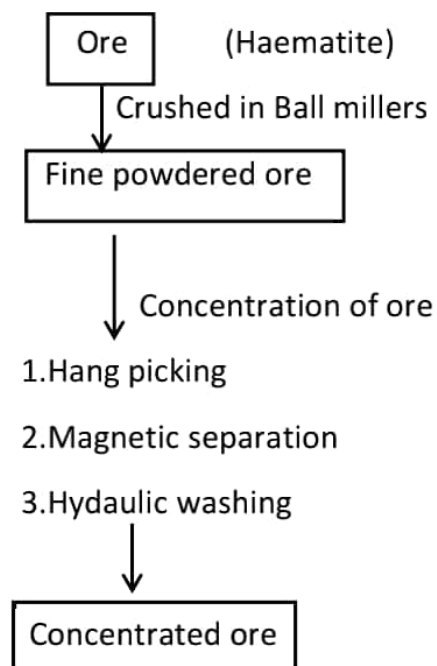


3. Extraction of Iron

Important minerals of Iron are

- 1) Haematite: Fe_2O_3
- 2) Iron pyrite: FeS_2
- 3) Siderite: FeCO_3
- 4) Limonite: $2 \text{Fe}_2\text{CO}_3 \cdot 2\text{H}_2\text{O}$
- 5) Magnetite: Fe_3O_4

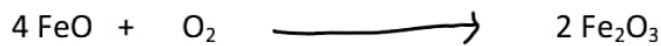
Iron mainly extracted from Haematite.



Concentrated Ore

Roasting

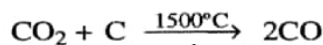
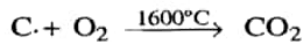
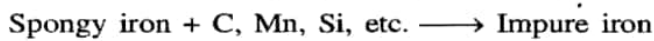
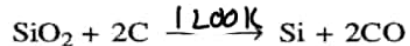
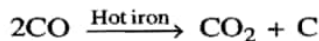
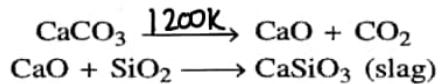
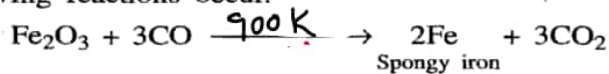
Volatile impurities like As, Sb, S are removed and their volatile oxides and also some amount of ferrous oxide converted to ferric oxide.



Roasted Ore

Smelting in blast furnace)

Smelting in a blast furnace ore + coke + limestone in 8:4:1 Ratio following reactions occur.



Pig iron

In Bessimer convertor

Sudden cooling

White cast iron

[contain carbon as cementite (Fe_3C)]

Slow cooling

grey cast iron

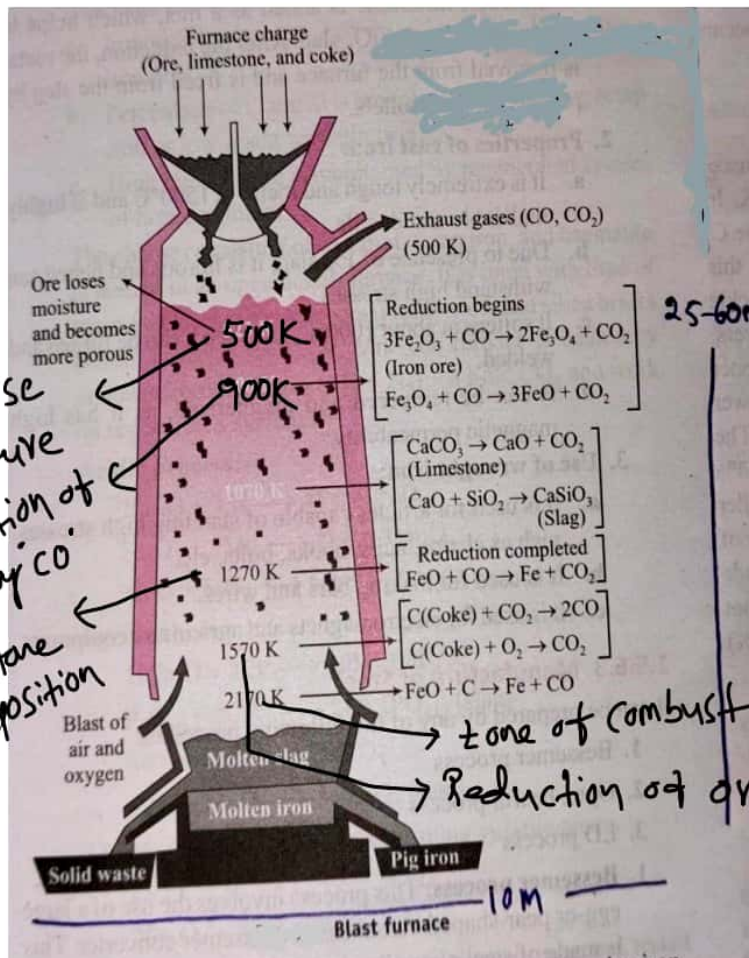
→ contain carbon as graphite

Cast iron

↓ Oxidation in reverberatory furnace

Wrought iron

Types of iron			
Property	Cast iron	Wrought iron	Steel
Chemical composition	Iron 93%-95% Carbon 2.5%-5% impurities about 2%	Iron 99.5%-99.8% Carbon 0.1%-0.21% impurities about 0.3%	Iron 98.0%-99.5% Carbon 0.25%-2.0%
Hardness	Very hard	Soft	Medium hardness
Magnetization	Cannot be permanently magnetized	Magnetization is not permanent but easy	Can be permanently magnetized
Malleability	Brittle	Malleable	Malleable and brittle
Melting point	Lowest about 1200°C	Highest about 1500°C	between 1300°C-1400°C
Structure	Crystalline	Fibrous	Granular



4. **Extraction of Aluminium:**

- It is highly reactive metal so doesn't occur in free state always occur in combine state.
- It is 1st most abundant metal & 3rd most abundant elements.

Important minerals:

1. Corundum: Al_2O_3
2. Diaspore: $\text{Al}_2\text{O}_3\text{H}_2\text{O}$
3. Bauxite: $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
4. Gibbsite: $\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
5. Cryolite: Na_3AlF_6
6. Feldspar: KAlSi_3O_8
7. Alunite: $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 4 \text{Al}(\text{OH})_3$
8. Mica: $\text{K}_2\text{O} \cdot 3\text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2 \cdot 2\text{H}_2\text{O}$

Aluminium mainly extracted from Bauxite using following steps

1. Purification of Bauxite to Alumina (Al_2O_3)

(By Baeyer's process, Hall's process and Serpeck's process)

2. Electrolytic Reduction of Alumina to crude Aluminium.

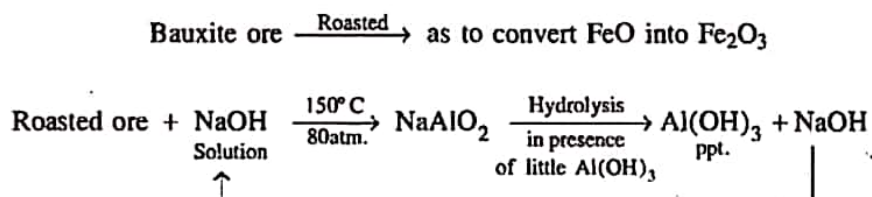
(By Hall's Heroult's Electrolytic Reduction)

3. Electrolytic Refining.

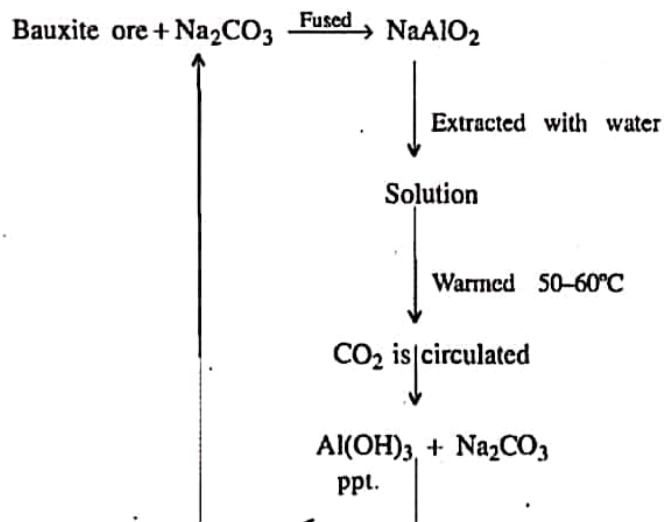
(By Hoops Electrolysis)

REFINING OF BAUXITE

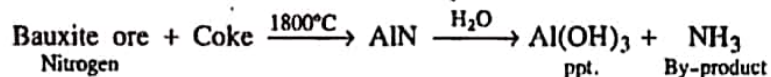
(a) Baeyer's Process :



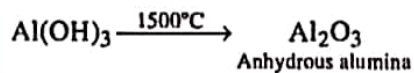
(b) Hall's Process :



(c) Serpeck's Process :



Calcination

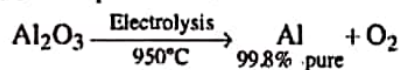


Electrolytic Reduction Hall and Heroult's Process

Electrolyte Al₂O₃ dissolved in Na₃AlF₆ and CaF₂

Cathode—Carbon lining

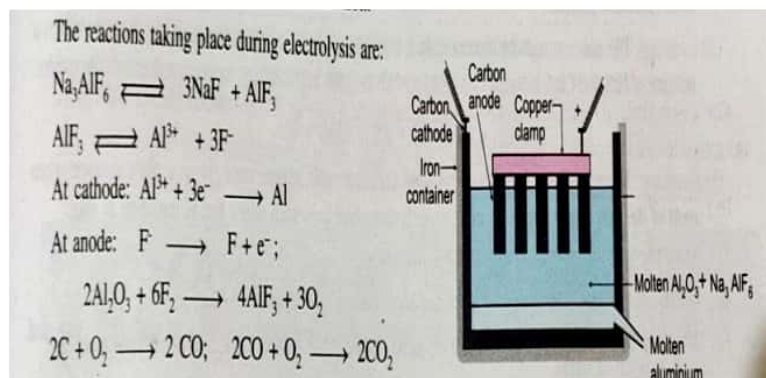
Anode—Graphite rods



Electrolytic Refining (Hoope's process)

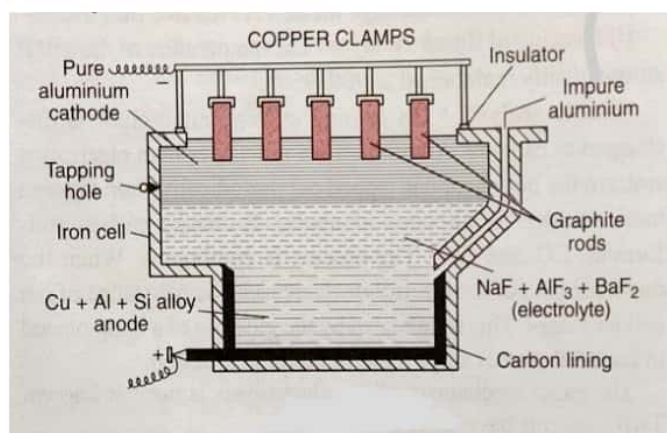
↓
Pure Al (99.98% pure)

Hall's Heroult's Electrolytic Reduction:



The oxygen liberated at anode reacts with carbon of the anode producing CO and CO₂. This way for each kg of Aluminium produced, about 0.5 kg carbon anode is burnt away. Aluminium set free at the cathode is removed periodically in the molten condition from the bottom of the cell.

Hoops Electrolysis:



Aluminium is refined by hope's electrolytic process. The refining cell consists of three fused layers of different densities. The bottom layer consists of impure Aluminium and forms an anode. The middle layer consists of a mixture of Cryolite and barium fluoride. The upper layer is of pure molten Aluminium which serves as the cathode.

On electrolysis Aluminium ions from the middle layer pass into upper layer and are discharged at the cathode as pure Aluminium metal. At the same time, an

equivalent amount of Aluminium from bottom layer goes into solution in the middle layer while the impurities remain below.

Important minerals of Silver:-

Extraction of Silver

① Silver glance (Argentite) $\Rightarrow \text{Ag}_2\text{S}$

② Horn silver $\Rightarrow \text{AgCl}$

③ Ruby silver $\Rightarrow \text{Ag}_3\text{SbS}_3$

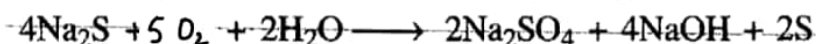
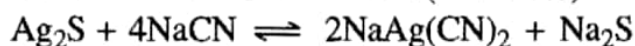
Silver ore
(Argentite)

CONCENTRATION BY FROTH FLOATATION PROCESS

Powdered ore + Water + Pine oil + Air $\longrightarrow$
Froth carrying sulphide particles

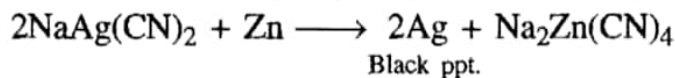
CYANIDATION

Concentrated ore + NaCN solution (0.4–0.6%) + Air



↓ Filtrate

PRECIPITATION OF SILVER WITH ZINC



Black ppt. of Ag + $\text{KNO}_3 \xrightarrow{\text{Fuse}}$ Compact mass
(Silver metal)

→ It contains impurities
like Au, Zn, Cu

↓

ELECTROLYTIC REFINING

Anode : Impure silver

Cathode : Pure silver plate

Electrolyte : AgNO_3 solution + HNO_3

Pure silver deposited on the cathode

- The process of desilverisation from lead is called Parke's process or Pattinson's reaction.

Extraction of Lead

Extraction of Lead:-

Important minerals:-

Galena:- PbS

Anglesite:- $PbSO_4$

Cerussite:- $PbCO_3$

Crocoisite:- $PbCrO_4$

Lanarkite:- $PbO \cdot PbSO_4$

[Ore] Galena.

↓ Crushed in
Ball miller.

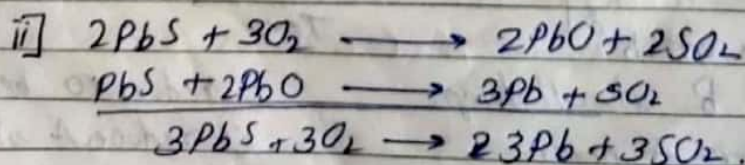
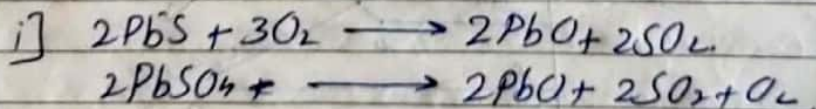
Fine powdered ore

↓ Concentration
of ore.

Froth Flotation with $NaCN$
to remove ZnS & SiO_2

↓ Concentrated ore.

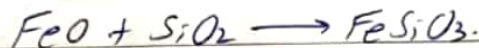
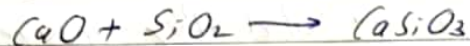
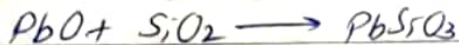
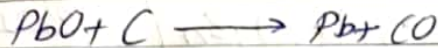
↓ Roasting.



Roasted ore.

Smelting

$PbO, PbSO_4, \text{Coke}, SiO_2, CaO, Fe_2O_3$



↓

crude Pb (Bi, Sb, Fe, Zn, Ni, Ag, Au)

in Air remove Bi, Sb

Softening of
crude lead.

Soft Lead

Desilverisation by parker's process.

↓

Desilverised Pb

Refining.

Bett's electrolysis.

Pure 'Pb' at cathode

Bett's electrolysis:- [For only refining of ^{Lead} metal]

Electrolyte:- $H_2SiF_6 + PbSiF_6 + \text{Gelatin}$

(Hexa Fluoro silicic acid) + lead Fluoro silicate

Anode:- Impure Pb

Cathode:- Pure Pb.

Extraction of Tin(Sn):

EXTRACTION OF TIN

Finely divided ore [SnO_2]
[cassiterite]

↓
Gravity separation
(Removes SiO_2 and other silicious matter)

↓
Concentrated ore
(containing SiO_2 in small amount)

↓
Roasting
(Removes SO_2 and As_2O_3)

↓
 SnO_2
(CuSO_4 , CuO , FeSO_4 , FeO (FeWO_4 + ... SiO_2)
(small amount)

↓
Washed with hot water,
(Removes CuSO_4 and FeSO_4)

↓
 SnO_2
(CuO + FeO) (FeWO_4 + ...) (SiO_2 small amount)

↓
Magnetic separation
(removes FeWO_4 + ...)

↓
Black tin
(SnO_2 75%)

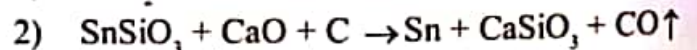
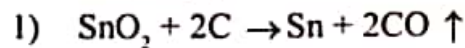
↓

(1) Carbon reduction
(anthracite coke + limestone Temp = 1200°C to 1300°C)

↓
Metallic Sn + slag (10-20% Sn)
(FeSiO_3 + SnSiO_3) (2)

↓
Smelted with coke + CaCO_3

↓
Metallic Sn



REFINING OF Sn:-

Crude Sn

(Iron, Cu oxides and tungsten are impurities)

↓
Liquation
(or sweating process)

↓
Pure Sn

↓
Poling
(Removes scum)

↓
99.5% pure Sn

↓
Electrorefining
(Electrolyte: SnSO_4 Cathode: Pure Sn Anode: Impure Sn)

↓
99.98% pure Sn

Ellingham Diagram:

Reduction of metal oxide using Thermodynamic process:

- Extraction of metals from their oxides by reduction using suitable reducing agent or applied high temperature is based on one thermodynamic property that is Gibbs free energy (G).

Gibbs free energy concept useful in selecting

- Suitable reducing agent
- Reaction possibility
- At what temperature reaction occur.

For any reaction or process Gibbs free energy change (ΔG) is given by the equation

$$\Delta G = \Delta H - T\Delta S$$

For the possibility of any reaction at any temperature ΔG must be negative.

CONDITIONS FOR ΔG TO BE NEGATIVE

ΔH	$T\Delta S$	Favourable conditions
- ve (Exothermic)	+ ve	Any temperature
- ve (Exothermic)	- ve	$\Delta H > T\Delta S$ (Temp. should be low)
+ ve (Endothermic)	+ ve	$T\Delta S > \Delta H$ (Temp. should be high)

- The graphical representation of Standard Gibbs energy change(ΔG°) Vs Temperature (T) was given by H.J.T Ellingham.
- The plot of free energy change, ΔG° against Temperature for the reaction of a metal and other elements with one mole of oxygen at 1 atmosphere OR Plots of G against TK are called the Ellingham diagrams.

1. Ellingham diagrams for the formation of metal oxides from metal:

$$2M_{(s)} + O_{2(g)} \rightarrow 2MO_{(s)} \quad (\Delta H = -ve)$$

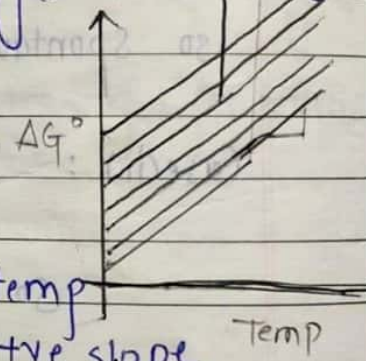
$\Rightarrow$ All combustion react^{ns} are exothermic.

$\Delta S = \text{No. of gaseous product} - \text{No. of gaseous reactants}$
 $\Delta S = 0 - 1 = -1$
 $\Delta S = -ve, \Delta H = -ve$

So $\Delta G^\circ = -ve$ at low temp &
 $\Delta G^\circ = +ve$ at high temp

ΔG° increase with increase in temp

$\Rightarrow$ Each graph is straight line with +ve slope
 $\rightarrow$ for few metal there is a sudden change for only
 ZnO, MgO, HgO (Phase change)



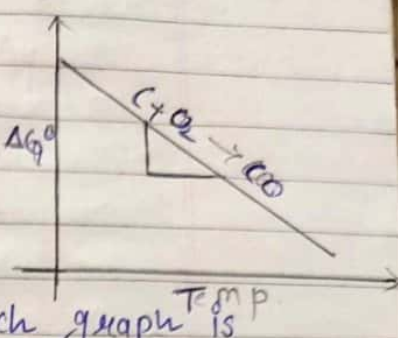
2. Ellingham diagrams for the formation of carbon monoxide from carbon

$$C_{(s)} + O_{2(g)} \rightarrow 2CO_{(g)} \quad \Delta H = -ve$$

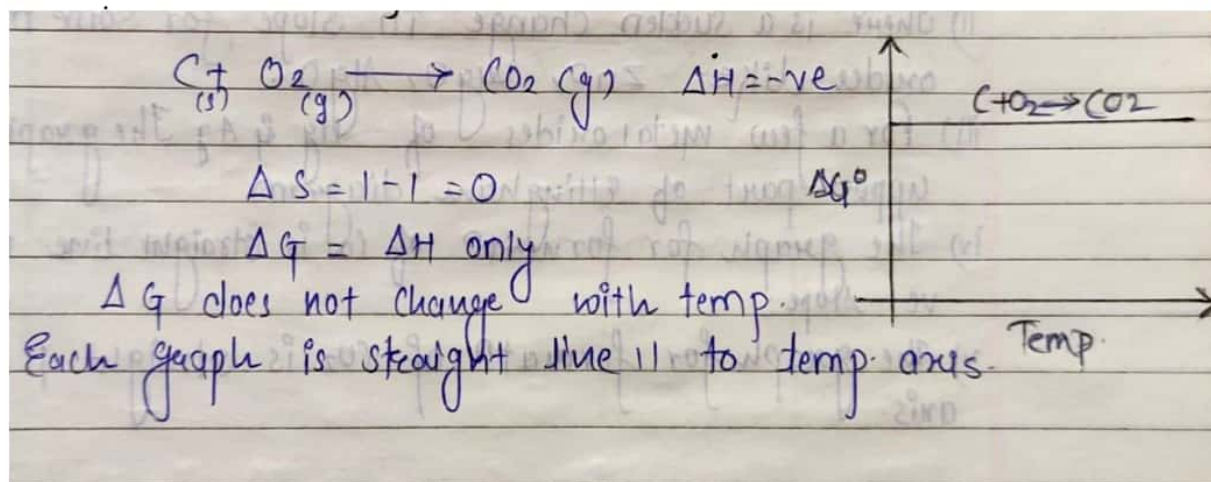
$\Delta S = 2 - 1 = 1 \quad \Delta S = +ve$

So, $\Delta H = -ve, \Delta S = +ve$,
 $\Delta G = -ve$ at Any temperature

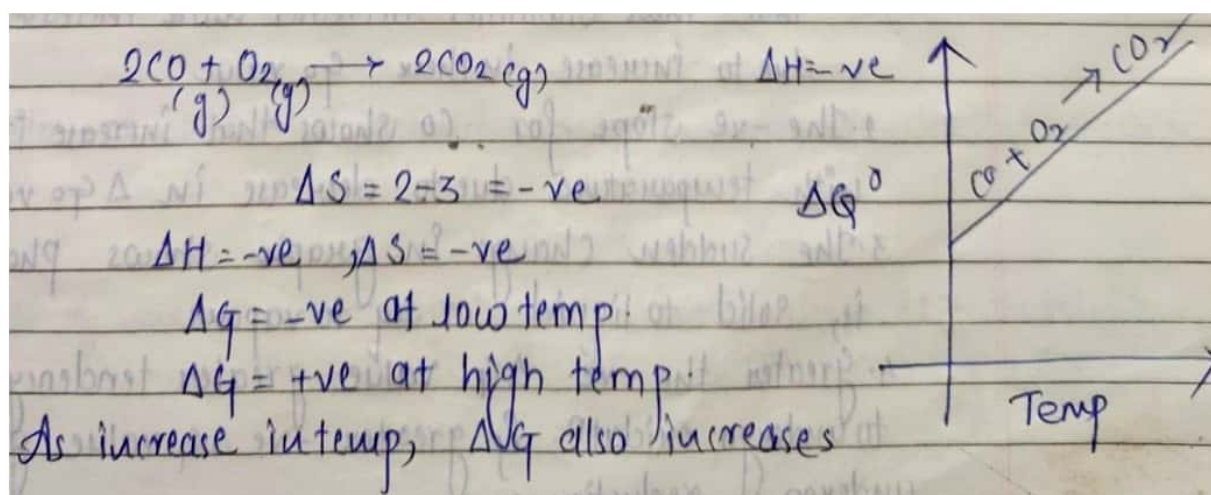
As increase in temp, ΔG decrease each graph is
 straight line with -ve slope

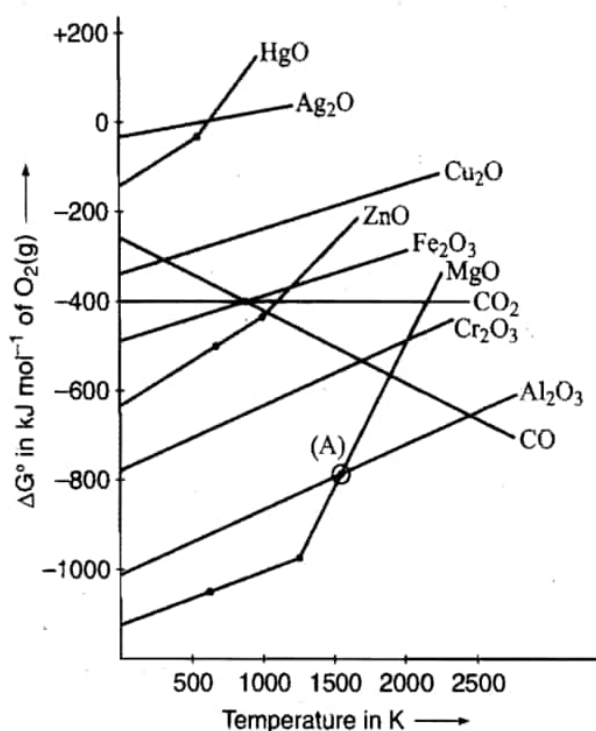


3. Ellingham diagrams for the formation of carbon dioxides from carbon:



4. Ellingham diagrams for the formation of carbon di oxides from carbon monoxide:





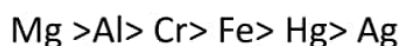
Ellingham diagrams for oxides of some elements

Features of Ellingham diagram:

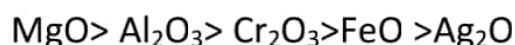
1. The graph for the formation of a metal oxide is a straight line with an upward slope.
2. There is sudden change in the slopes for some metal oxides, like MgO , ZnO and HgO .
3. For a few metal oxides of mercury and silver the graph is at the upper part in Ellingham diagrams (Ag_2O and HgO).
4. The graph for the formation of CO is straight line with negative slope. This line intersects the lines of many metal oxides.
5. The graph for the formation of CO_2 is straight line almost parallel to temperature axis.

Significance of Ellingham diagram:

1. The positive slope of metal oxides shows that their stabilities decrease with increase in temperature. The decrease in their stabilities is due to an increase in ΔG value.
2. The sudden change in the graph shows a phase change that is, change from solid to liquid or from liquid to vapour.
3. The negative slope of CO shows that it becomes more stable with increase in temperatures.
4. The relative tendency of the metals to undergo oxidation is in the order as shown below.

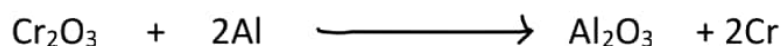


- This is due to the increase in the negative free energy change of the formation of oxide in the order given below



Selection of reducing agent:

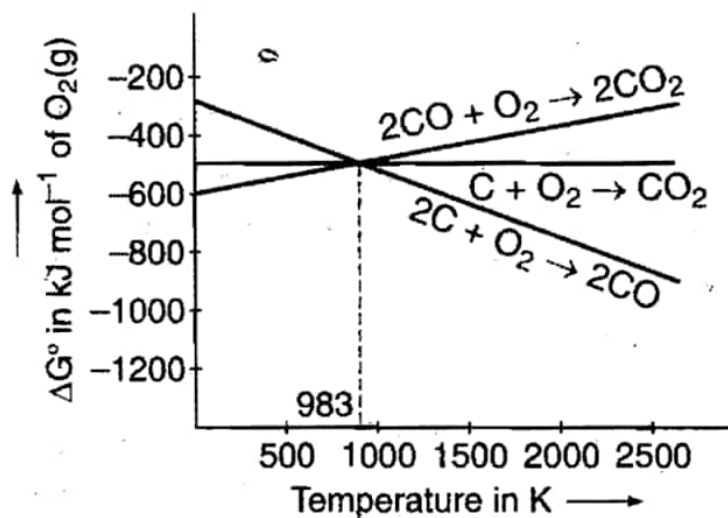
- Ellingham diagram tells us that any metal can reduce the oxide of other metal which appear above in the diagram.
- The graph for formation of oxides of Ag and Hg between ΔG & Temperature is at upper part of diagram, Because of their +VE ΔG value, so Ag_2O & HgO are unstable hence they can be decomposed by heating at moderate temperatures.
- Aluminium used as a reducing agent for Cr_2O_3 at all temperature because value of ΔG is more -VE for Al_2O_3 than Cr_2O_3 and both the curves parallel.

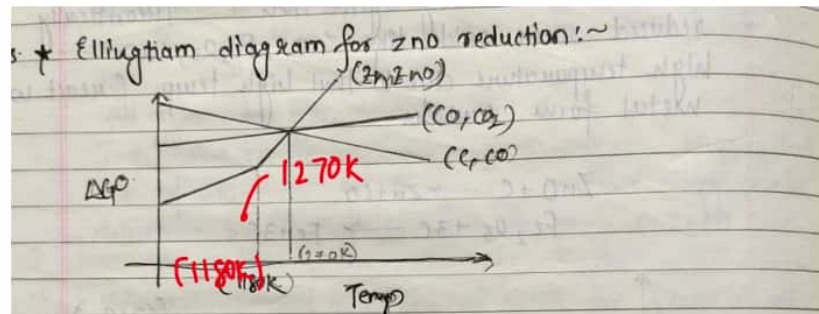


- In Ellingham diagram for MgO VS Al_2O_3 two curves at 1623K intersection is observed so below 1623K Mg reduces Al_2O_3 & above 1623K Al reduces MgO

Important of carbon & carbon monoxide as reducing agent:

- In Ellingham diagram formation of CO_2 from carbon & formation of CO from C intersects at 983K. It shows that CO is a better reducing agent at below 983K & at above 983K carbon acts as a better reducing agent. So practically found that CO is a better reducing agent than Carbon.
- Theoretically Carbon reduces some oxides like ZnO , Fe_2O_3 , SnO_2at high temperature due to that high temperature carbon reacts with metal to form carbides.

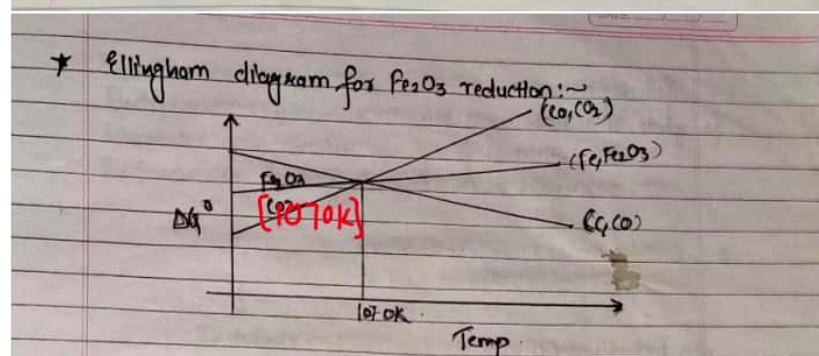




- ⇒ At 1180K Zn Boiled due to idication sudden increase the ΔG value.
- ⇒ At below 1270K Zn reduces CO to C

$$\text{Zn} + \text{CO} \rightarrow \text{ZnO} + \text{C}$$
- ⇒ At Above 1270K C reduces ZnO to Zn

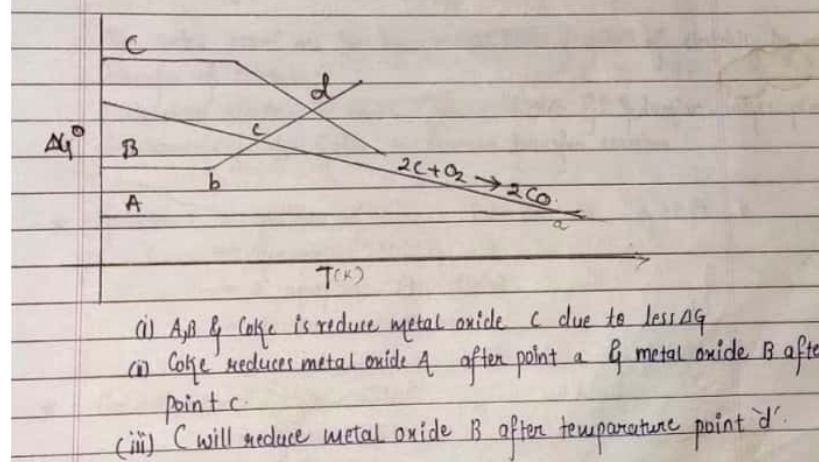
$$\text{ZnO} + \text{C} \xrightarrow{1673\text{K}} \text{Zn} + \text{CO}$$



- ⇒ At below 1070K CO reducing Fe_2O_3 to Fe

$$\text{Fe}_2\text{O}_3 + \text{CO} \xrightarrow{900\text{K}} 2\text{Fe} + 3\text{CO}_2$$
- ⇒ At above 1070K C reduces Fe_2O_3 to Fe

$$\text{Fe}_2\text{O}_3 + 3\text{C} \xrightarrow{1500\text{K}} 2\text{Fe} + 3\text{CO} \uparrow$$



Important Minerals

	Types of ore	Name of the ore	Composition of the ore
1.	Oxides	Corundum Diaspore Bauxite Tin stone Rutile Haematite Magnetite Limonite Chromite Cuprite (or ruby copper) Illmenite Zincite Pyrolusite	Al_2O_3 $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$ $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ SnO_2 TiO_2 Fe_2O_3 Fe_3O_4 $\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ $\text{FeO} \cdot \text{Cr}_2\text{O}_3$ Cu_2O $\text{FeO} \cdot \text{TiO}_2$ ZnO_2 MnO_2
2.	Sulphides	Galena Zinc blende Cinnabar Argentite (or silver glance) Pyrargyrite (or ruby silver) Iron pyrites Copper pyrites Chalcocite (or copper glance)	PbS ZnS HgS Ag_2S $3\text{Ag}_2\text{S} \cdot \text{Sb}_2\text{S}_2$ FeS_2 CuFeS_2 or $\text{Cu}_2\text{S} \cdot \text{Fe}_2\text{S}_3$ Cu_2S
3.	Halides	Rock salt Horn silver Carnallite Fluorspar Cryolite Sylvine	NaCl AgCl $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ CaF_2 Na_3AlF_6 KCl
4.	Carbonates	Magnesite Limestone Dolomite Calamine Malachite Azurite Cerrusite Siderite	MgCO_3 CaCO_3 $\text{CaCO}_3 \cdot \text{MgCO}_3$ ZnCO_3 $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ $\text{Cu}(\text{OH})_2 \cdot 2\text{CuCO}_3$ PbCO_3 FeCO_3

5.	Sulphates	Epsom salt or epsomite	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
		Barytes	BaSO_4
		Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
		Glauber's salt	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$
		Kainite	$\text{KCl} \cdot \text{MgSO}_4 \cdot 3\text{H}_2\text{O}$
		Calestine	SrSO_4
		Chalcanthite	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
		Anglesite	PbSO_4
		Anhydrite	CaSO_4
6.	Silicates	Petalite	$\text{LiAl}(\text{Si}_2\text{O}_5)_2$ (All are ores of Li)
		Beryl	$3\text{BeO} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$ (All are ores of beryllium)
		Willemite	Zn_2SiO_4
		China clay	$\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$
		Mica	$\text{K}_2\text{O} \cdot 3\text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2 \cdot 2\text{H}_2\text{O}$
		Felspar	KAlSi_3O_8 or $\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$
		Talc	$\text{Mg}_2(\text{Si}_2\text{O}_5) \cdot \text{Mg}(\text{OH})_2$
		Asbestos	$\text{CaMg}_3(\text{SiO}_3)_4$ or $\text{CaSiO}_3 \cdot 3\text{MgSiO}_3$
7.	Phosphates	Amblygonite	$\text{LiF} \cdot \text{AlPO}_4$
		Triphylite	$(\text{Li}, \text{Na})\text{PO}_4 \cdot (\text{Fe}, \text{Mn})_3(\text{PO}_4)_3$
		Chlorapatite	$\text{Ca}_3(\text{PO}_4)_3\text{Cl}$
		Fluorapatite	$3\text{Ca}_3(\text{PO}_4)_2\text{CaF}_2$
8.	Nitrates	Saltpetre	KNO_3
		Chile saltpetre	NaNO_3
9.	Arsenides	Kupfernickel	NiAs
		Nickel glance	NiAsS